PWPolyF: DSP-Based Piecewise-Polynomial Activations for Transformers on FPGAs

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MOTIVATION

Transformers apply nonlinear activations — **GELU, SiLU, Sigmoid, Tanh** — in every block. Vision Transformers are our motivating case.

FINN normally realises each as a **lookup table in BRAM / URAM** (thresholding).

When on-chip memory is tight, those tables compete with weights and attention.

PWPolyF adds **flexibility**: spend **DSPs instead of BRAM / URAM**, and pick the balance per design.

Scope: FINN currently supports only monotonically increasing activations — PWPolyF works within that set today (see Future Work).

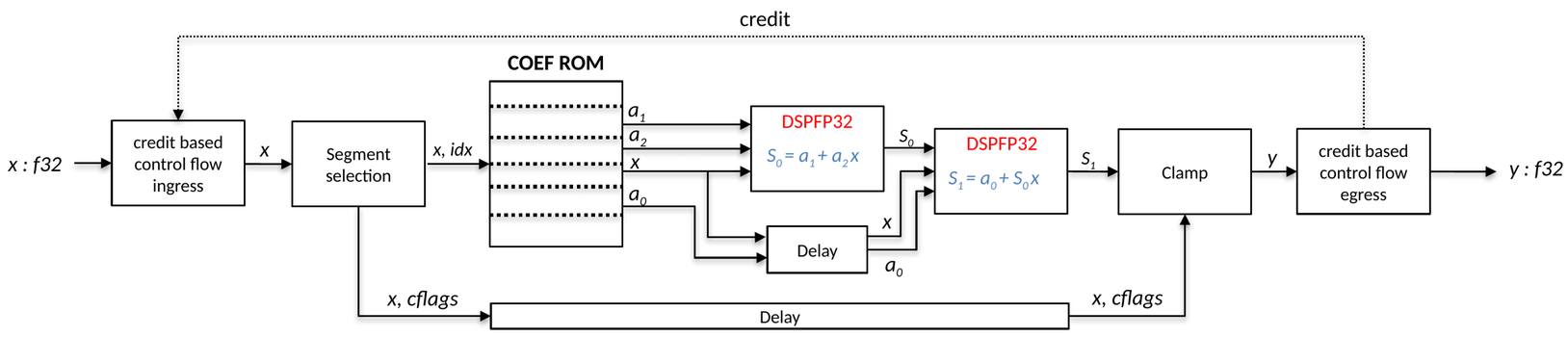
THE IDEA

- Replace the table with **math**: approximate each activation as **piecewise degree-2 polynomials**.
- Evaluate with **Horner's form a0 + x(a1 + a2 x)** — a short dependent FMA chain.
- Maps onto cascaded **DSP58 FP32 FMAs**: deeply pipelined, **high Fmax**, one result per cycle (II = 1), no BRAM.
- Works for any supported activation.



PWPolyF converts standard activation ops into a compact DSP approximation.

INSIDE PWPolyF · THE HORNER DATAPATH



x → segment select → coefficient ROM → two chained DSPFP32 FMAs (S0 = a1+a2·x, S1 = a0+S0·x) → clamp → y.

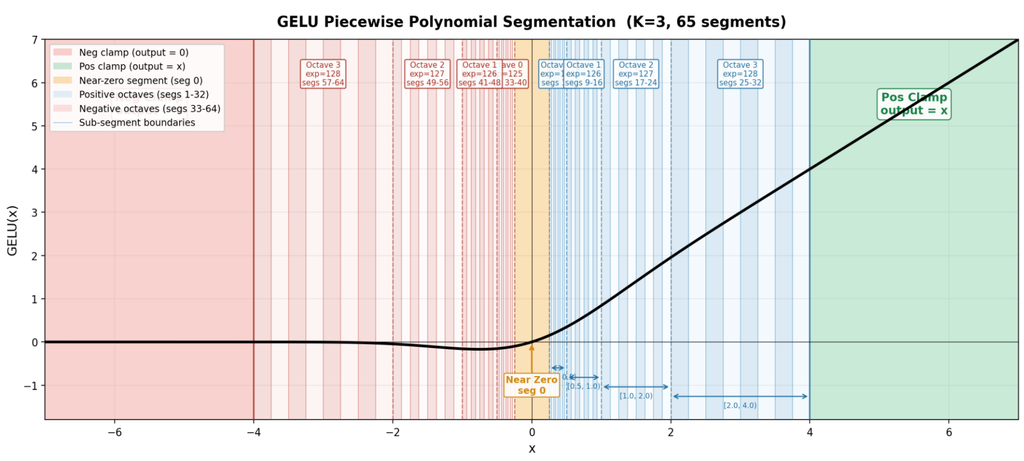
Requires AMD Versal

Both FMAs map onto the **DSP58** slice's hardened **FP32 multiply-add (DSPFP32)** — no soft-float logic, so the chain pipelines to high Fmax.

Builds on efficient DSP58 primitive sharing [1].

UltraScale+ DSP48 has no native FP32 FMA, so PWPolyF targets **Versal**.

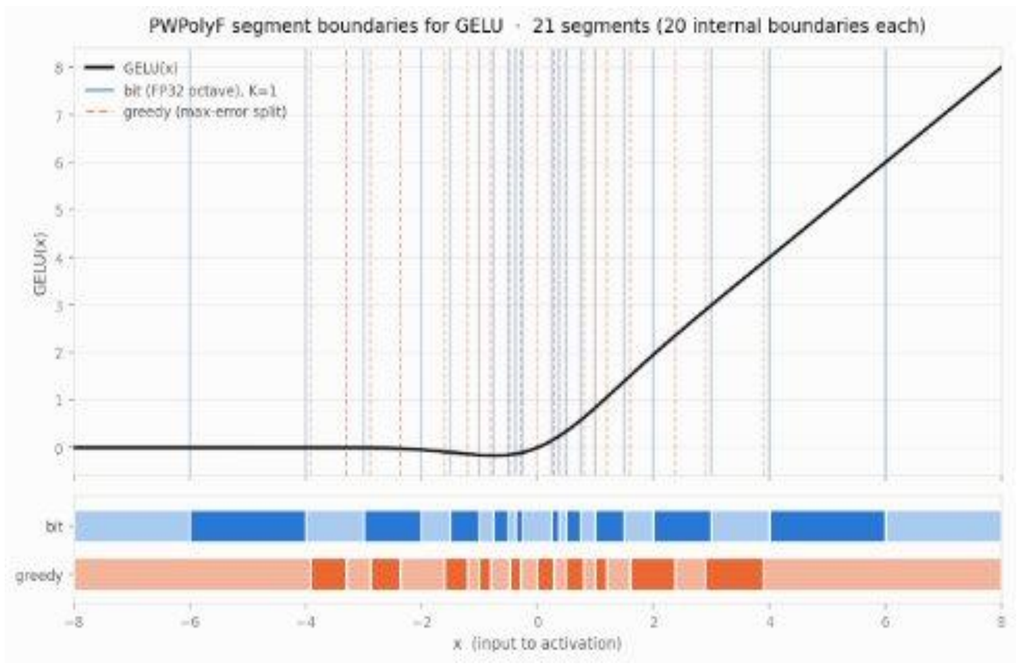
CHOOSING THE SEGMENTS



One polynomial per segment — dense where GELU bends, clamped in the flat tails.

Bit-extract

The FP32 exponent names the octave and the top-K mantissa bits subdivide it, so segment select is pure **bit-slicing — near-zero extra LUTs**. Power-of-2 boundaries; reuses the FMA/DSP58 sharing of [1].



Threshold

Compare x against a sorted FP32 boundary list (**torch.bucketize** / fp32_le): N thresholds → N+1 segments at (segs−1)×lanes LUTs. **Only segment-select changes — the FMA datapath is identical.** Boundaries can match the bit edges, be uniform, or be placed by a **greedy error-driven partitioner** that splits where the polynomial fit is worst. Coefficients refit per segment.

RESULTS (per lane, II = 1)

Partition	K	LUT	DSP	Fmax (MHz)
Bit-extract	2	256	2	410.7
Bit-extract	3	339	2	360.0
Bit-extract	4	412	2	370.4
Threshold	3	390	2	347.0

Per activation lane on AMD Versal — 2 DSP58 and 0 BRAM across all configs; only LUTs and Fmax vary.

CONTRIBUTIONS

- **PR #1573** New RTL PWPolyF activation; **InferPWPolyFLayer** auto-converts standard ONNX activations; folded by SetFolding.
- **PR #1629** Adds threshold partitioning (**partitionMode**) with user-defined boundaries.

WHY IT MATTERS

- Trades BRAM / URAM tables for a compact DSP datapath — **frees on-chip memory** for weights and attention.
- Single dependent FMA chain on hardened DSP58 FP32 — deeply pipelined, high Fmax, II = 1.
- Bit-extract for production; threshold to move accuracy where the function needs it.

FUTURE WORK

- Support non-monotonic activations in FINN.
- Automate boundary selection from accuracy / resource targets.
- Fold K and boundary choice into the folding / resource search.

[1] *FMA Is (Almost) All You Need! Sharing FMA Resources for Linear and Non-linear Function Acceleration;* Arpan Suravi Prasad , Andrea Belano, Davide Rossi, Francesco Conti , Luca Benini; ARITH 2026.